

PRECAUTION

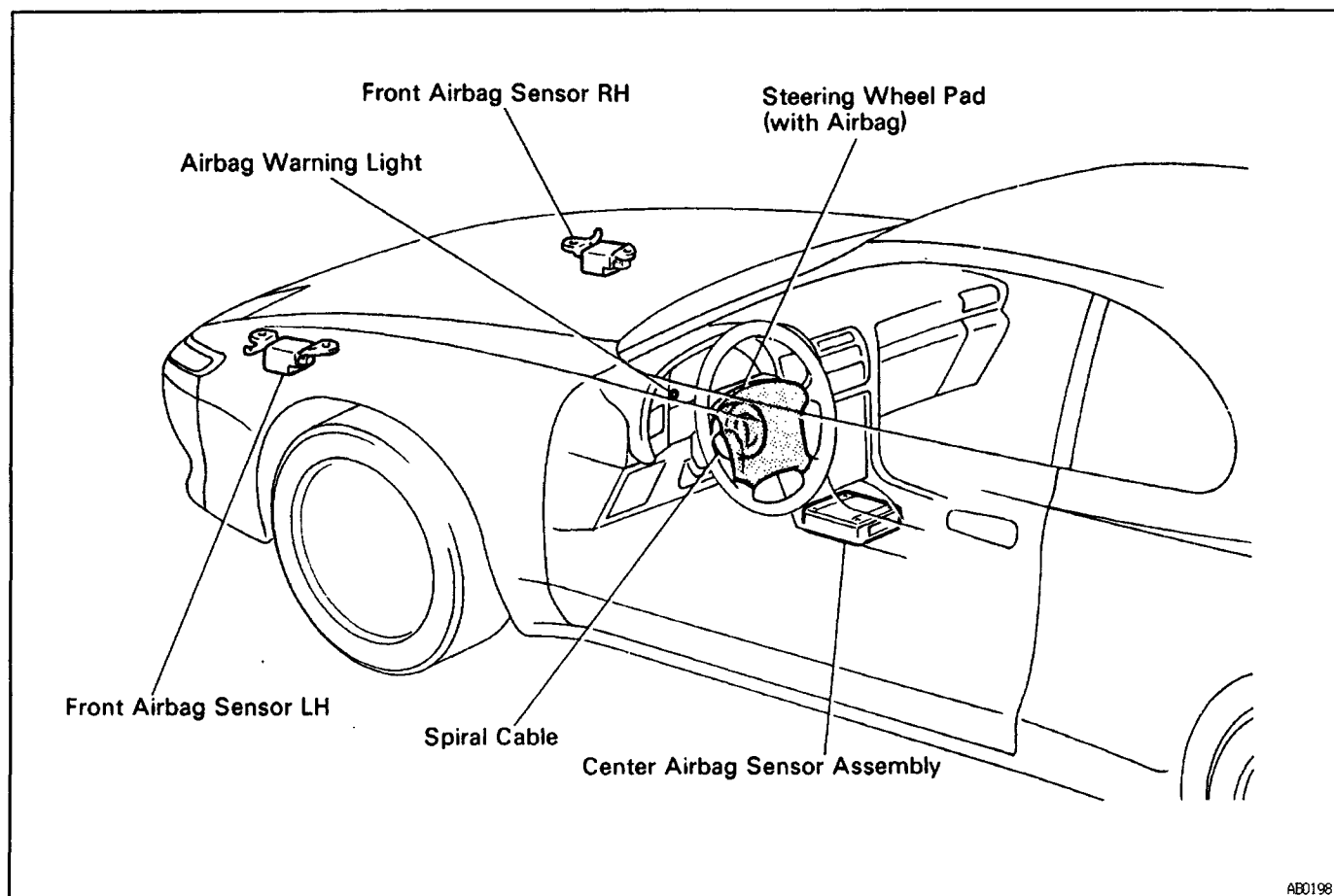
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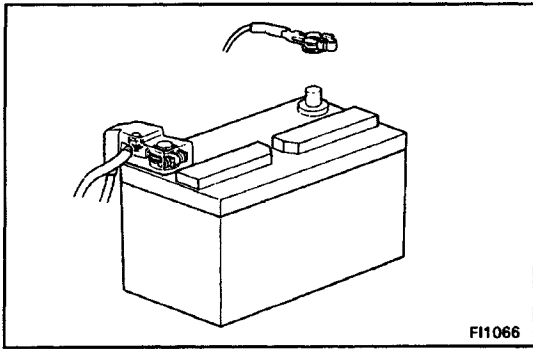
FOR VEHICLES EQUIPPED WITH SRS AIRBAG

The 1993 CELICA for USA specifications is equipped with an SRS (Supplemental Restraint System) airbag. Failure to carry out service operations in the correct sequence could cause the airbag system to unexpectedly deploy during servicing, possibly leading to a serious accident.

Further, if a mistake is made in servicing the airbag system, it is possible the airbag may fail to operate when required. Before performing servicing (including removal or installation of parts, inspection or replacement), be sure to read the following items carefully, then follow the correct procedure described in this manual.

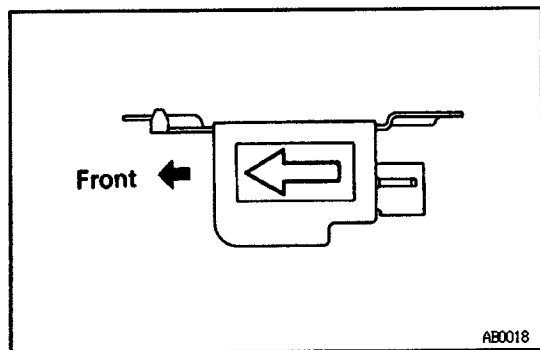
Locations of Airbag Components





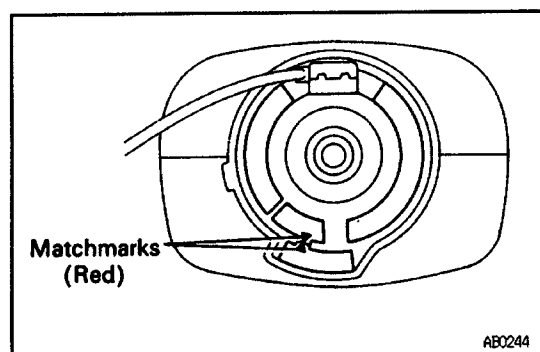
1. Malfunction symptoms of the airbag system are difficult to confirm, so the diagnostic codes become the most important source of information when troubleshooting. When troubleshooting the airbag system, always inspect the diagnostic codes before disconnecting the battery (See page [AB-24](#)).
2. **Work must be started after approx. 20 seconds or longer from the time the ignition switch is turned to the LOCK position and the negative (-) terminal cable is disconnected from the battery.**
(The airbag system is equipped with a back-up power source so that if work is started within 20 seconds of disconnecting the negative (-) terminal cable of the battery, the airbag may be deployed.)
When the negative (-) terminal cable is disconnected from the battery, memory of the clock and audio systems will be cancelled. So before starting work, make a record of the contents memorized by each memory system. Then when work is finished, reset the clock and audio systems as before.
To avoid erasing the memory of each memory system, never use a back-up power supply from outside the vehicle.
3. Even in cases of a minor collision where the airbag does not deploy, the front airbag sensors and the steering wheel pad should be inspected (See page [AB-11](#)).
4. Never use airbag parts from another vehicle. When replacing parts, replace them with new parts.
5. Before repairs, remove the airbag sensors if shocks are likely to be applied to the sensors during repairs.
6. The center airbag sensor assembly contains mercury. After performing replacement, do not destroy the old part. When scrapping the vehicle or replacing the center airbag sensor assembly itself, remove the center airbag sensor assembly and dispose of it as toxic waste.
7. Never disassemble and repair the front airbag sensors, center airbag sensor assembly or steering wheel pad in order to reuse it.
8. If the front airbag sensors, center airbag sensor assembly or steering wheel pad have been dropped, or if there are cracks, dents or other defects in the case, bracket or connector, replace them with new ones.
9. Do not expose the front airbag sensors, center airbag sensor assembly or steering wheel pad directly to hot air or flames.
10. Use a volt/ohmmeter with high impedance (10 k Ω /V minimum) for troubleshooting of the electrical circuit.

11. Information labels are attached to the periphery of the airbag components. Follow the notices.
12. After work on the airbag system is completed, perform the airbag warning light check (See page AB-29).



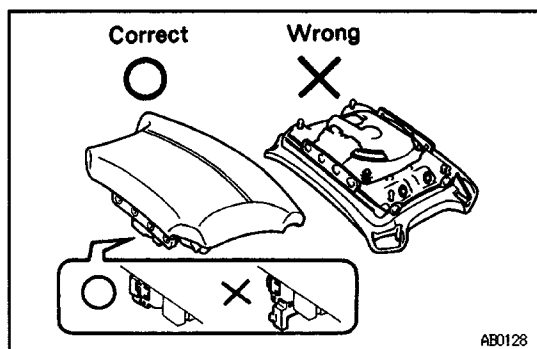
Front Airbag Sensor

1. Never reuse the front airbag sensors involved in a collision when the airbag has deployed. (Replace both left and right airbag sensors.)
2. Install the front airbag sensor with the arrow on the sensor facing toward the front of the vehicle.
3. The front airbag sensor set bolts have been anti-rust treated. When the sensor is removed, always replace the set bolts with new ones.
4. The front airbag sensor is equipped with an electrical connection check mechanism. Be sure to lock this mechanism securely when connecting the connector. If the connector is not securely locked, a malfunction code will be detected by the diagnosis system (See page AB-9).



Spiral Cable (in Combination Switch)

The steering wheel must be fitted correctly to the steering column with the spiral cable at the neutral position; otherwise cable disconnection and other troubles may result. Refer to page AB-16 concerning correct steering wheel installation.

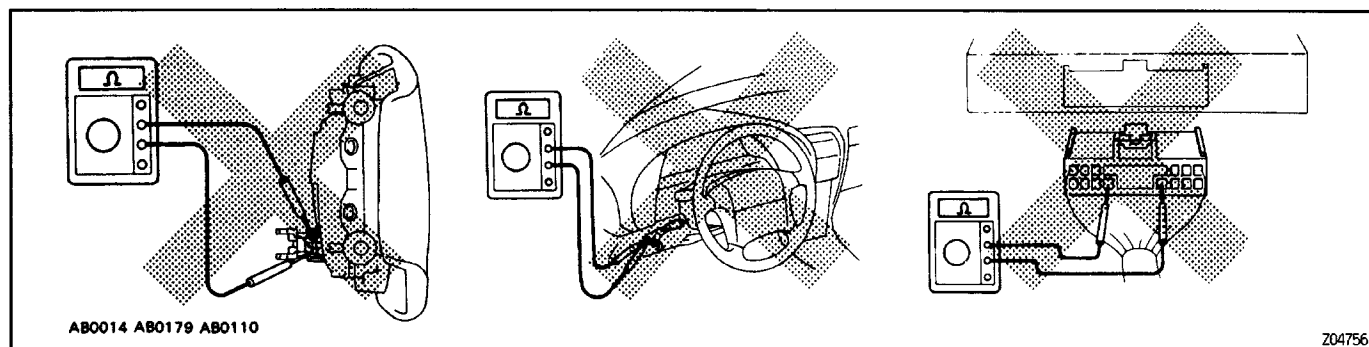


Steering Wheel Pad (with Airbag)

1. When removing the steering wheel pad or handling a new steering wheel pad, it should be placed with the pad top surface facing up.

In this case, the twin-lock type connector lock lever should be in the locked state and care should be taken to place it so the connector will not be damaged. And do not store a steering wheel pad on top of another one. (Storing the pad with its metallic surface up may lead to a serious accident if the airbag inflates for some reason.)

2. Never measure the resistance of the airbag squib. (This may cause the airbag to deploy, which is very dangerous.)



3. Grease should not be applied to the steering wheel pad and the pad should not be cleaned with detergents of any kind.
4. Store the steering wheel pad where the ambient temperature remains below 93°C (200°F), without high humidity and away from electrical noise.
5. When using electric welding, first disconnect the airbag connector (yellow color and 2 pins) under the steering column near the combination switch connector before starting work.
6. When disposing of a vehicle or the steering wheel pad alone, the airbag should be deployed using an SST before disposal (See page AB-82). Perform the operation in a place away from electrical noise.

Center Airbag Sensor Assembly

The connector to the center airbag sensor assembly should be connected or disconnected with the sensor mounted on the floor. If the connector is connected or disconnected while the center airbag sensor assembly is not mounted to the floor, it could cause undesired ignition of the airbag system.

Wire Harness and Connector

The airbag system's wire harness is integrated with the cowl wire harness assembly. The wires for the airbag wire harness are encased in a yellow corrugated tube. All the connectors for the system are also a standard yellow color. If the airbag system wire harness becomes disconnected or the connector becomes broken due to an accident, etc., repair or replace it as shown on page [AB-21](#).

FOR VEHICLES EQUIPPED WITH A CATALYTIC CONVERTER

CAUTION: If large amounts of unburned gasoline flow into the converter, it may overheat and create a fire hazard. To prevent this, observe the following precautions and explain them to your customer.

1. Use only unleaded gasoline.

2. Avoid prolonged idling.

Avoid running the engine at idle speed for more than 20 minutes.

3. Avoid spark jump test.

(a) Perform spark jump test only when absolutely necessary. Perform this test as rapidly as possible.

(b) While testing, never race the engine.

4. Avoid prolonged engine compression measurement.

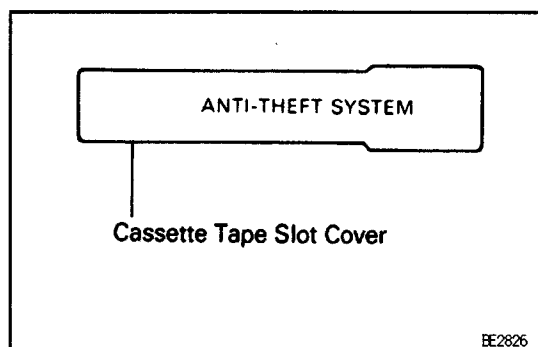
Engine compression tests must be done as rapidly as possible.

5. Do not run engine when fuel tank is nearly empty.

This may cause the engine to misfire and create an extra load on the converter.

6. Avoid coasting with ignition turned off and prolonged braking.

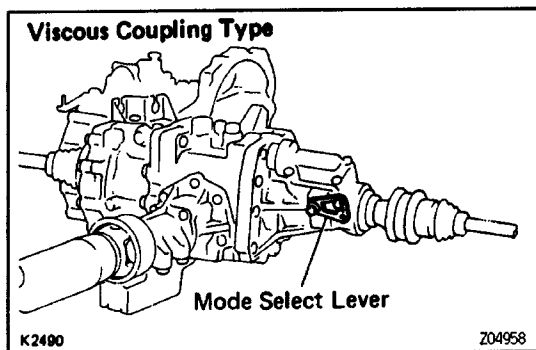
7. Do not dispose of used catalyst along with parts contaminated with gasoline or oil.



FOR VEHICLES WITH AN AUDIO SYSTEM WITH BUILT-IN ANTI-THEFT SYSTEM

Audio System displaying the sign "ANTI – THEFT SYSTEM" shown on the left has a built-in anti-theft system which makes the audio system soundless if stolen.

If the power source for the audio system is cut even once, the anti-theft system operates so that even if the power source is reconnected, the audio system will not produce any sound unless the ID number selected by the customer is input again. Accordingly, when performing repairs on vehicles equipped with this system, before disconnecting the battery terminals or removing the audio system the customer should be asked for the ID number so that the technician can input the ID number afterwards, or else a request made to the customer to input the ID number. For the method to input the ID number or cancel the anti-theft system, refer to the Owner's Manual.



WHEN SERVICING FULL-TIME 4WD VEHICLES

IN020-01

The full-time 4WD Celica is equipped with the viscous coupling type of center differential lock. When carrying out any kind of servicing or testing on a full-time 4WD in which the front or rear wheels are made to rotate (braking test, speedometer test, on-the-car wheel balancing, etc.), or when towing the vehicle, be sure to observe the precautions given below. If incorrect preparations or test procedures are used, the test will not be able to be successfully carried out, and may be dangerous as well. Therefore, before beginning any such servicing or test, be sure to check the following items:

- (1) Center differential lock type
- (2) Center differential mode position
- (3) Whether wheels should be touching ground or jacked up
- (4) Transmission gear position
- (5) Maximum testing vehicle speed
- (6) Maximum testing time

Also be sure to observe the following cautions:

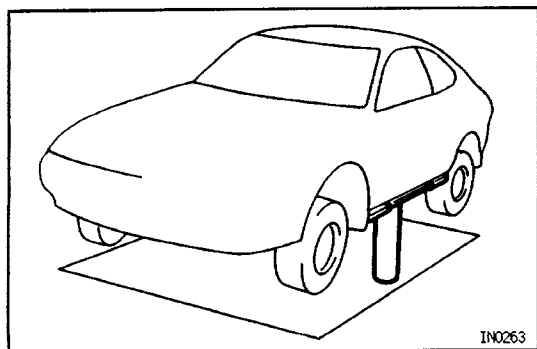
- (1) Never accelerate or decelerate the vehicle suddenly.
- (2) Observe the other cautions given for each individual test.

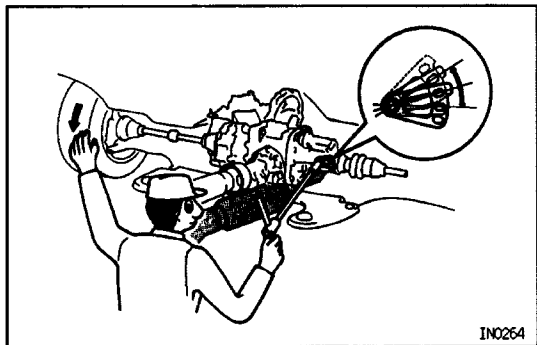
Before Beginning Test

During tests with a brake tester or chassis dynamometer, such as braking force tests or speedometer tests, if only the front or the rear wheels are to be rotated, it is necessary to set the Mode Select Lever on the transaxle to the Viscous Free Mode or to the FF Mode depending on the type of test being performed. In addition, after moving the lever to the position of the desired mode, be sure to check that the center differential's state has changed accordingly.

Moving Mode Select Lever

1. JACK UP VEHICLE





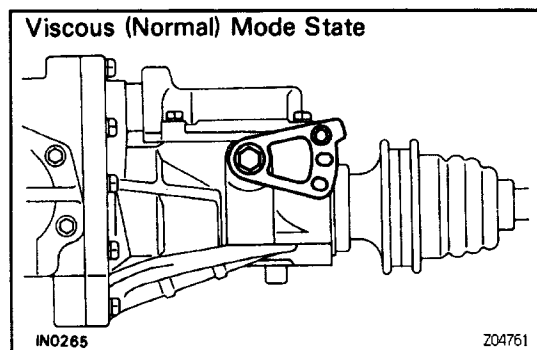
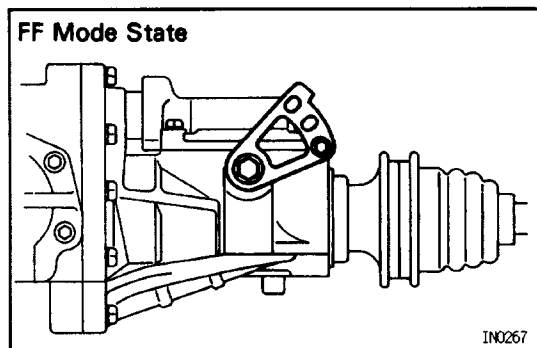
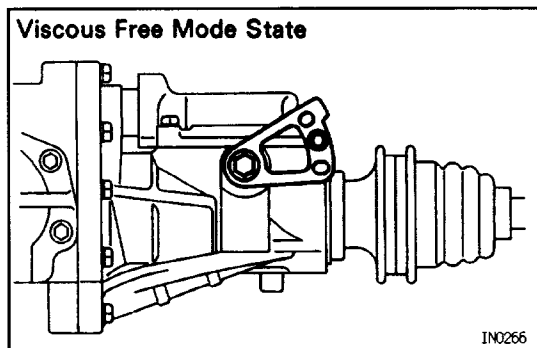
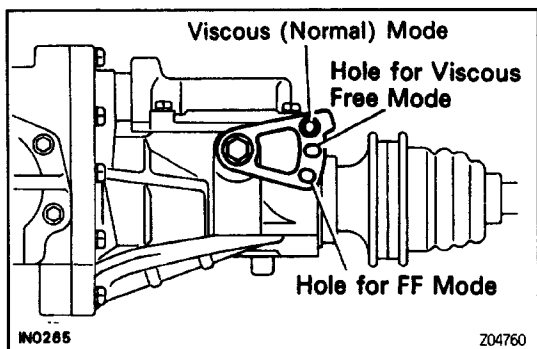
2. REMOVE MODE SELECT LEVER SET BOLT

3. MOVE MODE SELECT LEVER TO DESIRED MODE POSITION

HINT:

1. If the mode select lever cannot be moved smoothly, shift the transmission to 1st gear, then move the lever while rotating one front wheel by hand.
2. Do not use excessive force when moving the mode select lever.

4. LOCK MODE SELECT LEVER WITH SET BOLT



5. CONFIRM MODE

Viscous Free Mode:

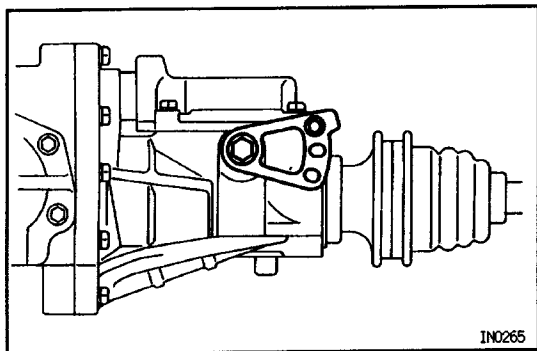
Jack up one of the front wheels and check that the wheel can be rotated by hand with the transmission in neutral.

FF Mode:

Jack up one of the rear wheels and check that the wheel can be rotated by hand with the transmission in 1st gear.

Viscous (Normal) Mode:

Jack up one of the front wheels and check that the wheel resists being rotated by hand with the transmission in neutral.



6. AFTER FINISHING TEST, RETURN MODE SELECT LEVER TO "VISCOUS (NORMAL) MODE" POSITION AND INSTALL SET BOLT

NOTICE:

1. After moving the mode select lever, jack up one of the front or rear wheels and check that the wheel rotates to confirm that the mode selection has been made correctly.
2. Be sure to tighten the set bolt securely each time after moving the mode select lever.
3. Do not engage the clutch or pump the accelerator or brakes suddenly in the viscous free mode or the FF mode.
4. If either the front or the rear wheels are placed on the tester rollers in the viscous free mode, be careful not to exceed the following limits:

Maximum speed:

Speed indicated on speedometer

30 km/h (19 mph) or

Wheel speed (tester speed)

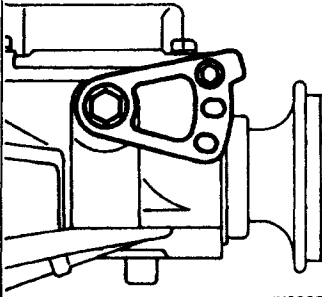
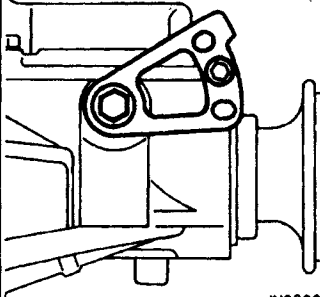
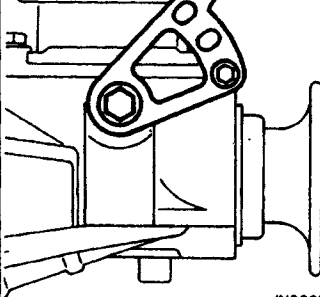
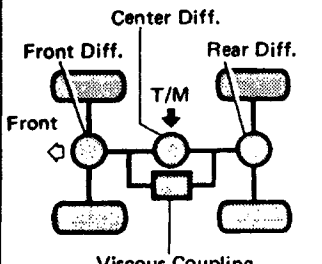
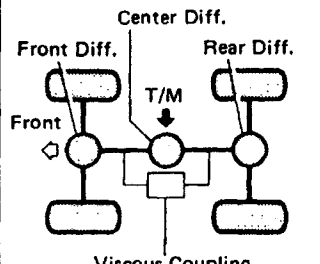
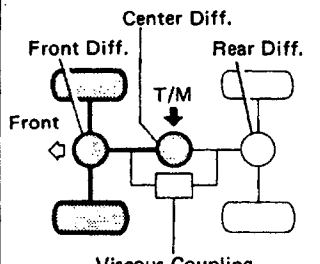
60 km/h (38 mph)

Maximum test time: 60 sec.

Note that the actual wheel speed (tester speed) is twice the speed indicated by the speedometer due to center differential operation.

5. Do not drive the vehicle in 1 st gear, 2nd gear or in reverse while in FF mode. If it is necessary to move the vehicle, drive it in 3rd, 4th or 5th gear. When desiring to back the vehicle, push it backwards manually.
6. After finishing the test, be sure to move the mode select lever back to the viscous (normal) mode and lock it securely with the set bolt.

State in Each Mode

Mode	Viscous (Normal) Mode	Viscous Free Mode	FF Mode
Mode Select Lever Position	 IN0265	 IN0266	 IN0267
State in Each Mode	 CP0052	 CP0053	 CP0054
Viscous Coupling	Operating	Not Operating	Not Operating
Center Differential	Free	Free	Locked and Separated from Rear Drive
Driving Wheel:	4WD	4WD	FWD
Conditions of Use	Normal Driving	When using a brake tester	- When using a chassis dynamometer - When using a combination tester
		Never use this mode during normal driving	

Braking Force Test

HINT:

1. According to the vehicle speed during the test, select one of the two test methods described below, either A or B.
2. The mode select lever position of mode select lever differs in the two test methods, A and B, so take adequate precautions.
Method A (Low Speed Test): Viscous Free Mode
Method B (High Speed Test): FF Mode

Test Method A (Low Speed Test)

Speed indicated on speedometer:

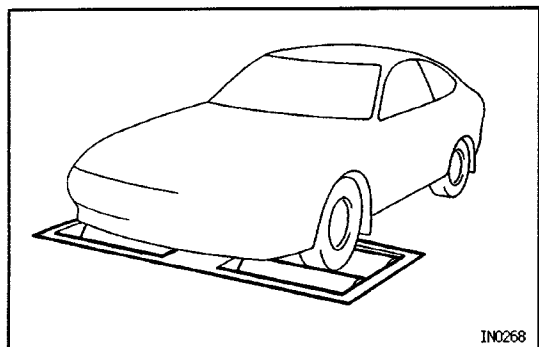
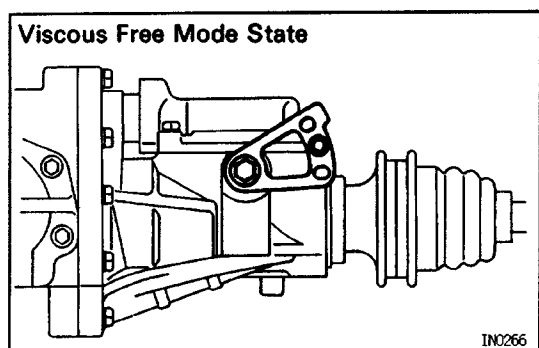
Below 30 km/h (19 mph),

Wheel speed (tester speed):

Below 60 km/h (38 mph) and

Test time:

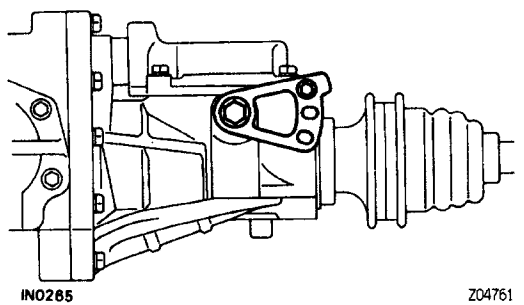
Within 60 sec.



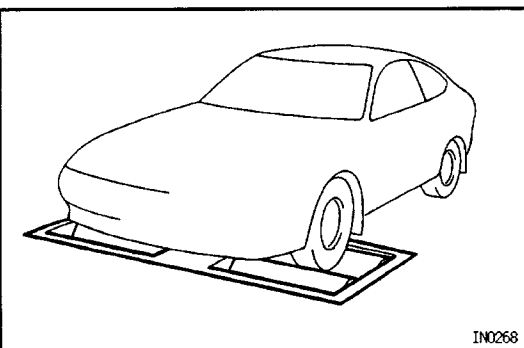
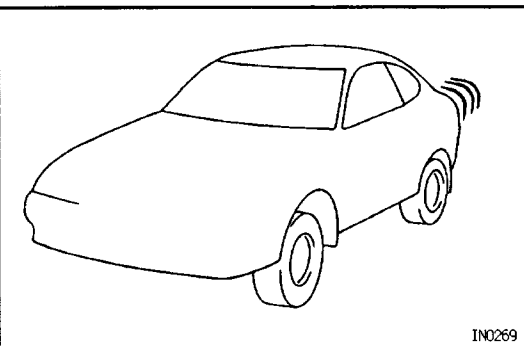
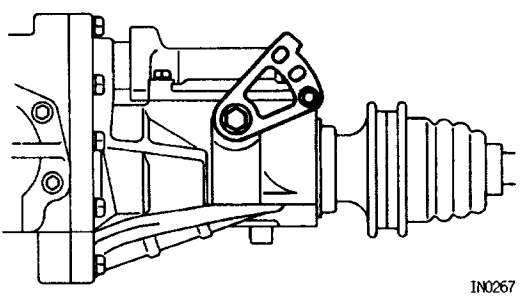
1. MOVE MODE SELECT LEVER ON TRANSAXLE TO "VISCIOUS FREE MODE" AND INSTALL SET BOLT
(See page [IN-15](#))
2. CONFIRM MODE SELECTION
(See page [IN-16](#))
3. PLACE WHEELS (EITHER FRONT OR REAR) ON TESTER ROLLERS
HINT: The actual wheel speed (tester speed) is twice the speed indicated by the speedometer due –to center differential operation, so take adequate precautions.
4. DISCONNECT INERTIA WEIGHT FROM TESTER ROLLER (If equipped with it)
5. PUT TRANSMISSION IN NEUTRAL
6. OPERATE TESTER ROLLERS AND MEASURE BRAKING FORCE .

HINT: Since different types of tester are used, such as specialized brake testers and combination testers with built – in chassis dynamometer, speedometer tester, brake tester, etc., conduct the test in accordance with the instructions furnished for the tester model used.

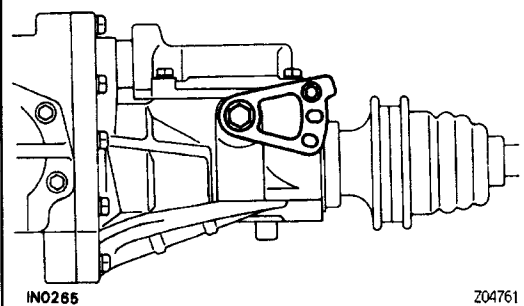
Viscous (Normal) Mode State



FF Mode State



Viscous (Normal) Mode State



7. AFTER FINISHING TEST, BE SURE TO MOVE MODE SELECT LEVER BACK TO "VISCIOUS (NORMAL) MODE" AND LOCK IT SECURELY WITH SET BOLT

Test method B (High Speed Test)

[Vehicle Speed: Over 60 km/h (38 mph)]

1. MOVE MODE SELECT LEVER TO "FF MODE" AND INSTALL SET BOLT
(See page [IN-15](#))
2. CONFIRM MODE SELECTION
(See page [IN-16](#))
3. PLACE WHEELS (EITHER FRONT OR REAR) TO BE TESTED ON TESTER ROLLERS

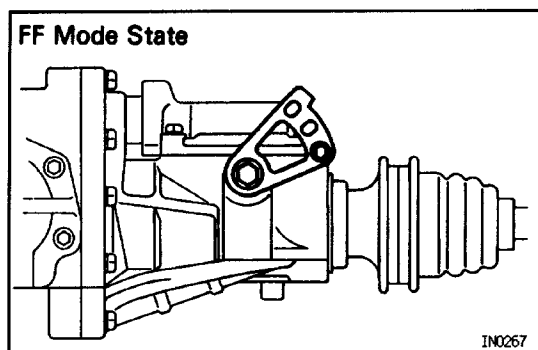
NOTICE:

1. Do not drive the vehicle in 1 st gear, 2nd gear or reverse while in the FF mode.
2. If it is necessary to move the vehicle, drive it in 3rd, 4th or 5th gear.
3. When desiring to back the vehicle, push it back-wards manually.
4. Do not engage the clutch or pump the accelerator or brake pedal suddenly.

4. DISCONNECT INERTIA WEIGHT FROM TESTER ROLLER (if equipped with it)
5. PUT TRANSMISSION IN NEUTRAL
6. OPERATE TESTER ROLLERS AND MEASURE BRAKING FORCE

Tester operation differs depending on the type of tester used. Be sure to follow the procedure specified in the instructions supplied by the tester manufacturer.

7. AFTER FINISHING TEST, BE SURE TO MOVE MODE SELECT LEVER BACK TO " VISCIOUS (NORMAL) MODE" AND LOCK IT SECURELY WITH SET BOLT



Speedometer Test or Other Tests (Using Speedometer Tester or Chassis Dynamometer)

1. MOVE MODE SELECT LEVER TO "FF MODE" AND
INSTALL SET BOLT

(See page [IN-15](#))

2. CONFIRM MODE SELECTION

(See page [IN-16](#))

NOTICE:

1. Do not drive the vehicle in 1st gear, 2nd gear or reverse while in the FF mode.
2. If it is necessary to move the vehicle, drive it in 3rd, 4th or 5th gear.
3. When desiring to back the vehicle, push it backwards manually.
4. Do not engage the clutch or pump the accelerator or brake pedal suddenly.

3. PLACE FRONT WHEELS ON TESTER ROLLERS

4. CHECK REAR WHEELS

5. APPLY PARKING BRAKE

6. DISCONNECT INERTIA WEIGHT FROM TESTER
ROLLER (if equipped with it)

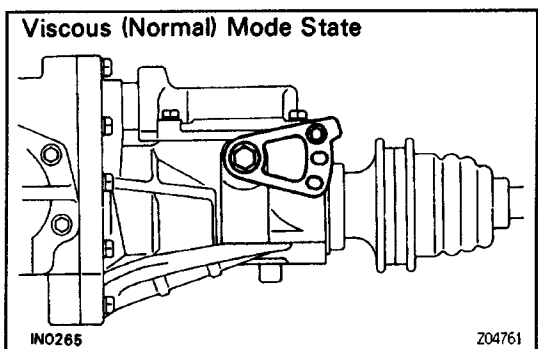
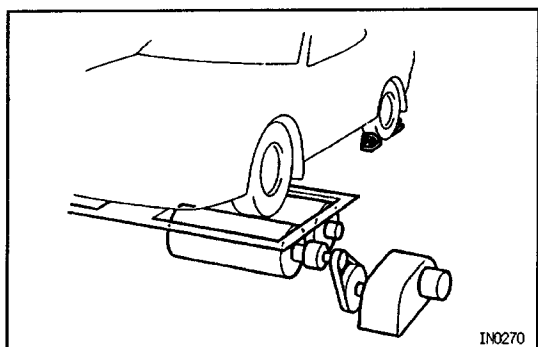
7. TEST VEHICLE

- (1) Start the engine.
- (2) Put the transmission in 3rd gear.
- (3) Engage the clutch slowly, then gradually increase the speed as the test is conducted.

HINT: The test should be conducted in 3rd, 4th and 5th gears.

- (4) After the test is finished, reduce the speed gradually, then stop the engine.

8. AFTER FINISHING TEST, BE SURE TO MOVE MODE
SELECT LEVER BACK TO "VISCOUS (NORMAL)
MODE" AND LOCK IT SECURELY WITH SET BOLT

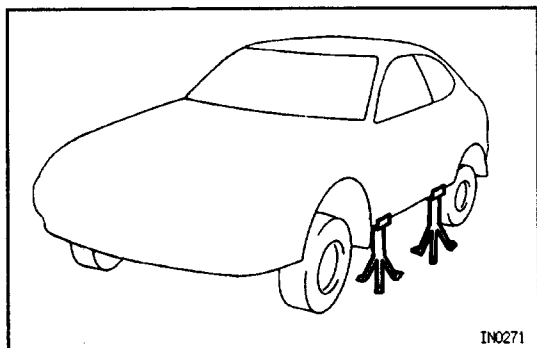


On-The –Car Wheel Balancing

NOTICE:

1. When doing on-the-car wheel balancing on a full-time 4WD vehicle, to prevent the wheels from rotating at different speeds on indifferent directions from each other (which could lead to damage to the center differential or transaxle gears), always be sure to observe the following precautions:

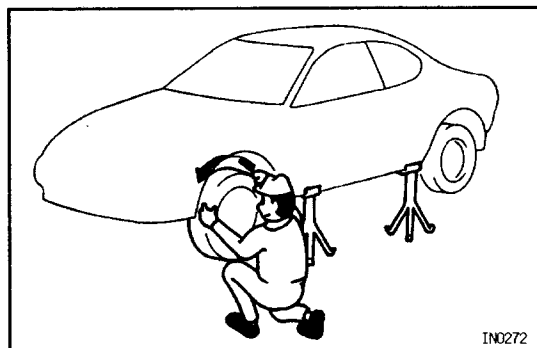
- (a) All four wheels should be jacked up, clearing the ground completely.
 - (b) The wheels should be driven with both the engine and the wheel balancer.
 - (e) The mode select lever on the transaxle of the viscous coupling type center differential should be in the viscous (normal) mode position.
 - (d) The parking brake lever should be fully released.
 - (e) None of the brakes should be allowed to drag.
2. Avoid sudden acceleration, deceleration and braking.
 3. Carry out the wheel balancing with the transmission in 3rd or 4th gear.



1. JACK UP VEHICLE SO THAT ALL FOUR WHEELS CLEAR GROUND AND CAN BE ROTATED

The wheels be rotating fast, so make sure the vehicle is firmly supported on stands.

2. RELEASE PARKING BRAKE FULLY

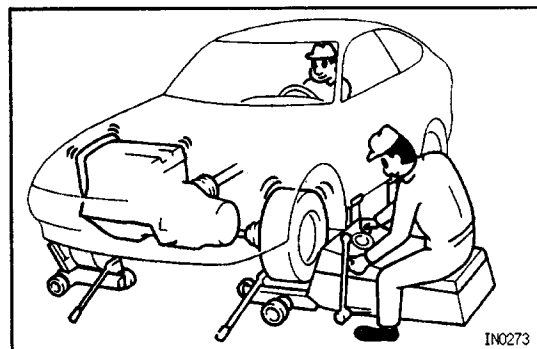


3. MAKE SURE THAT BRAKES ARE NOT DRAGGING ON ANY OF FOUR WHEELS

4. PLACE WHEEL TO BE BALANCED ON WHEEL BALANCER

Follow the procedure specified by the wheel balancer–manufacturer.

5. START ENGINE



6. PUT TRANSMISSION IN 3RD OR 4TH GEAR

7. ENGAGE CLUTCH SLOWLY, THEN GRADUALLY INCREASE SPEED TO TEST SPEED

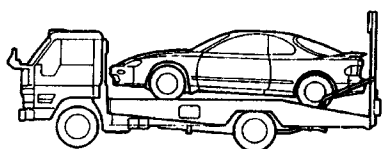
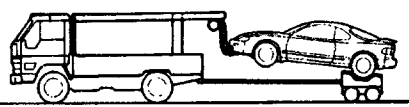
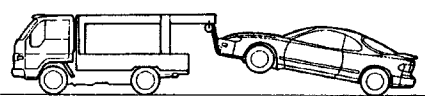
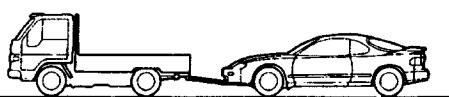
8. ROTATE WHEELS USING BOTH ENGINE'S DRIVING FORCE AND DRIVING FORCE OF WHEEL BALANCER AND CHECK WHEEL BALANCE

HINT: When doing this, be careful of the other wheels which will rotate at the same time.

WHEN TOWING FULL-TIME 4WD VEHICLES

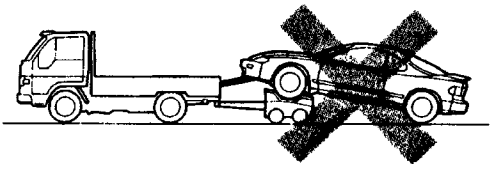
IN00Y-01

1. Use one of the methods shown below to tow the vehicle.
2. When there is trouble with the chassis and drive train, use method (1) (flat bed truck) or method (2) (sling type tow truck with dollies)
3. Recommended Methods: No. (1), (2), (3)
- Emergency Method: No. (4)

Towing Method \ Type of Center Differential	Viscous Coupling Type		
	Parking Brake	T/M Shift Lever Position	Viscous Coupling
① Flat Bed Truck  IN0274	Applied	1st Gear	Normal Driving Viscous Mode)
② Sling-Type Tow Truck with Dollies  IN0275			(No Special Operation Necessary)
③ Sling-Type Tow Truck (Front wheels must be able to rotate freely)  IN0276	Released	Neutral	↑
④ Towing with a Rope  IN0277	Released	Neutral	↑

HINT: Do not use any towing method other than those shown above.

For example, the towing method shown below is dangerous, so do not use it.

 IN0278	During towing with this towing method, there is a danger of the drivetrain heating up and causing breakdown, or of the front wheels flying off the dolly.
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